भारतीय प्रौद्योगिकी संस्थान जोधपुर

Indian Institute of Technology Jodhpur

Concept Note for MSc in Computational Social Sciences (MSc CSS)

School of Liberal Arts (SoLA)
Indian Institute of Technology Jodhpur

1. BACKGROUND

The Master of Science in Computational Social Science (MSc CSS) offered by the School of Liberal Arts (SoLA) at the Indian Institute of Technology Jodhpur was launched in 2022 to address the growing need for interdisciplinary training that integrates social science inquiry with computational methods and data-driven approaches. Since its introduction, the field of Computational Social Science has evolved rapidly, driven by advances in artificial intelligence, machine learning, digital data systems, and large-scale societal data analysis. In addition, institutional academic guidelines and credit requirements have been revised, necessitating periodic review and update of existing academic programmes.

In this context, the school proposes a revision of the MSc CSS curriculum based on revised institute credit requirements, feedback from current and former students, inputs from faculty members who have previously taught in the programme, and the evolving academic resources within SoLA. The revision seeks to strengthen the interdisciplinary orientation of the programme, update course content, rationalise the credit structure, introduce courses in emerging areas, and enhance opportunities for research, electives, and industry-academia engagement. This revision will help maintain the programme's academic relevance and future orientation.

Eligibility

- 3 Yr or 4 Yr UG Degree, with a First class in any subject of Social Sciences, Humanities, Basic Science, Management and all engineering branches.
- Mathematics up to Class XII is desirable.

Non-Graded (NG) Course and Thesis

- As per the new guidelines, students will undertake a 2-credit Non-Graded (NG) course on Industry-Academia Summer Internship.
- Students will undertake their thesis in the 3rd Semester, with the work continuing into the 4th Semester.

Course Level

- Students may opt for the required number of Programme Electives from the approved list.
- Students may also opt for either 600-level or 700-level courses as Open Electives.

2. OBJECTIVES

- To examine traditional social science questions using data-driven technologies, computational methods, and digital tools.
- To equip students with the knowledge and skills required to formulate, analyse, and address social science questions through technology-enabled approaches.
- To enable students to understand complex and rapidly changing social realities through transdisciplinary perspectives.
- To establish a distinctive academic programme that prepares future professionals and responsible citizens to develop innovative solutions for societal challenges.

The MSc in Computational Social Science introduces students to the emerging field, which employs computational methods, digital technologies, and data-driven approaches to study dynamic social processes and systems. The programme seeks to develop socially informed and technologically proficient professionals who can critically understand social structures, inequalities, and transformations while contributing innovative solutions for a sustainable and equitable future.

3. GRADUATE ATTRIBUTES

At the end of the programme, graduates are expected to:

- Demonstrate a transdisciplinary perspective for addressing social science questions through computational methods, data-driven tools, and digital technologies.
- Possess a strong theoretical understanding of social systems and proficiency in managing, analysing, and interpreting societal data for real-world applications.
- Demonstrate expertise in the visualisation and computational analysis of social data to support evidence-based decision-making and professional practice.
- Exhibit a research-oriented mindset that fosters critical inquiry, innovation, and interdisciplinary engagement in the investigation of social phenomena.

4. LEARNING OUTCOMES

Upon successful completion of the MSc in Computational Social Science programme, graduates will be able to:

- Integrate theories, concepts, and perspectives from the social sciences, humanities, and computational sciences to analyse complex social phenomena from a transdisciplinary perspective.
- Apply computational, statistical, and data-driven techniques to collect, manage, analyse, visualise, and interpret social data for addressing contemporary societal challenges.
- Design and conduct ethically informed and methodologically rigorous research using qualitative, quantitative, and computational approaches.
- Develop innovative technology-enabled solutions and communicate analytical insights effectively for applications in research, policy, industry, and public engagement.

5. SEMESTER-WISE DISTRIBUTION OF CREDITS

Sl No	Semester	Course Category	Credit	Total Credit
1	I	Programme Compulsory (PC)	18	21
2		Lab (PC)	3	
3	II	Programme Compulsory (PC)	18	19
4		Lab (PC)	1	
5	Summer Semester	Internship (NG)	2	2
6	III	Programme Elective (PE)	9	15
7		Open Elective (OE)	3	
8		Thesis	3	
9	IV	Programme Elective (PE)	3	15
10		Open Elective (OE)	3	
11		Thesis	9	
Total				**72**

Credit Distribution

Sl No	Particular	Total Credit
1	Programme Compulsory (PC)	40
2	Programme Elective (PE)	12
3	Open Elective (OE)	6
4	Thesis	12
5	NG	2
Total		**72**

6. PROGRAMME STRUCTURE

Semester-I				
Type	**Level**	**Name**	**LTP**	**Credit**
PC	600	Fundamentals of Computational Social Sciences	(3-0-0)	3
PC	600	Sociology of Science and Technology	(3-0-0)	3
PC	600	Ethics for Computational Social Sciences	(3-0-0)	3
PC	600	Language, Society and Computation	(3-0-0)	3
PC	600	Social Science Research Methods	(3-0-0)	3
PC	600	Computational Methods for Social Sciences	(3-0-0)	3
PC	600	Computational Social Sciences Lab	(0-0-2)	1
PC	600	Statistics Lab	(0-0-4)	2
Total Credit				**21**

Semester-II				
Type	**Level**	**Name**	**LTP**	**Credit**
PC	600	Computational Methods in Indian Knowledge Systems	(3-0-0)	3
PC	600	Health and Society	(3-0-0)	3
PC	600	Introduction to Policy Studies	(3-0-0)	3
PC	600	Social Realities	(3-0-0)	3
PC	600	Applied Economic Foundations for Social Sciences	(3-0-0)	3
PC	600	AI and Machine Learning for Social Sciences & Humanities	(3-0-0)	3
PC	600	AI & ML Lab	(0-0-2)	1
Total Credit				**19**

Summer				
Type	**Level**	**Name**	**LTP**	**Credit**
NG	NG	Internship		2
Total Credit				**2**

Semester-III				
Type	**Level**	**Name**	**LTP**	**Credit**
PE	700	Programme Elective-1	(3-0-0)	3
PE	700	Programme Elective-2	(3-0-0)	3
PE	700	Programme Elective-3	(3-0-0)	3
OE	600/700	Open Elective-1	(3-0-0)	3
T	700	Thesis	(0-0-3)	3
Total Credit				**15**

Semester-IV				
Type	**Level**	**Name**	**LTP**	**Credit**

PE	700	Programme Elective-4	(3-0-0)	3
OE	600/700	Open Elective-2	(3-0-0)	3
T	700	Thesis	(0-0-9)	9
Total Credit				**15**

##Learners of the MSc CSS will undertake the Thesis under the supervision of the faculty of SoLA. If needed, they may take joint supervision from another Department/ School/Centre of IIT Jodhpur.

7. PROGRAMME ELECTIVE

Sl No	Level	Course name	LTP	Credit
1	700	Theoretical Foundations of Econometrics	3-0-0	3
2	700	Economic Challenges and Data-Driven Approach	3-0-0	3
3	700	Global Health	3-0-0	3
4	700	Health Policy, Systems and Financing	3-0-0	3
5	700	Historical and Social Anthropology	3-0-0	3
6	700	Sociology of Education	3-0-0	3
7	700	Politics of Water Governance	3-0-0	3
8	700	Cognition & Emotion	3-0-0	3
9	700	Human Behaviour in Context: Neuro-cultural and Digital Perspectives	3-0-0	3
10	700	Contemporary Western Philosophy	3-0-0	3
11	700	Indian Knowledge Systems (IKS)	3-0-0	3
12	700	Sentence Structure	3-0-0	3

First Year Compulsory Courses

Title	**Fundamentals of Computational Social Sciences**	**Number**	LAL6050
Department	SoLA	L-T-P [C]	3-0-0[3]
Offered for	MSc	Type	Core for MSc CSS, Elective for others
Prerequisite	Nil		

Objectives

1. This course examines how knowledge and research are generated and used in the field of social sciences.
2. To understand major issues in epistemology and the philosophy of science concerning what we can know about the world, including a discussion of discipline-based and interdisciplinary approaches.
3. Explores basic principles of research theory, with attention to the advantages and disadvantages of research designs such as aggregate analysis, small-n comparisons, and case studies.
4. To appreciate the role of computational methods in social sciences.

Learning Outcomes

1. To understand how knowledge and research are generated and used in the field of social sciences.
2. Learn about major issues in epistemology and the philosophy of science concerning what we can know about the world, including a discussion of discipline-based and interdisciplinary approaches.
3. Understand different methods of social inquiry.
4. To understand the role of computational methods in the social sciences.

Contents

Introduction to epistemology: Theories of knowledge and the role of disciplinary approaches to it; history and philosophy of science; advantages and disadvantages of interdisciplinary approaches; ambiguities of knowledge generation and the uncertainties that surround attempts to know what we know and better (6 lectures)

Philosophy of science and the aims of social science: Nature of knowledge, theory, and evidence in social Investigation; key steps in 'science' progression; claims about the nature of the world that social scientists make; nature and scope of major social science disciplines; Hierarchy of Science, Social Science Challenges (6 lectures).

Developing and testing theories: Role of theory generation and theory testing in social research; diffusion theory, agenda-setting theory, and institutional political economy versus interest group theory. (7 lectures)

Marrying computational methods with social science: History of CSS; common grounds between computational methods and social science; fourth paradigm: A 21st century science; digital revolution. (7 lectures)

Role of computational methods in social sciences: Computational methods in social science; regression methods for establishing causal relations and their limitations; computational methods in social policies; secondary datasets and their relevance in social science research; history, essence and future of agent-based modelling; computer simulation in solving mysteries of social emergence (11 lectures).

Evolution of CSS field: Difference between computational social science and conventional social science; emergence of new computational tools and methods; opportunities and challenges in CSS (2 lectures).

Textbook

- Kuhn, Thomas. 1970. The Structure of Scientific Revolutions. Chicago: University of Chicago Press, 2nd edition. ISBN: 0226458121.

- King, Gary, Robert O. Keohane, and Sidney Verba. 1994. Designing Social Inquiry. Princeton: Princeton University Press. ISBN: 0691034710.
- Booth, Wayne C. 2008. The Craft of Research, 3rd edition. Chicago: University of Chicago Press. ISBN-10: 0226065669.

References

- Salganik, M. J. (2019). Bit by bit: Social research in the digital age. Princeton University Press.
- Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabási, A. L., Brewer, D., ... & Van Alstyne, M. (2009). Computational Social Science. Science, 323(5915), 721-723.
- Conte, R., Gilbert, N., Bonelli, G., Cioffi-Revilla, C., Deffuant, G., Kertesz, J., ... & Helbing, D. (2012). Manifesto of computational social science. The European Physical Journal Special Topics, 214(1), 325-346

Online Resource:

https://www.coursera.org/learn/computational-social-science-methods/home/welcome

Title	Sociology of Science and Technology	Number	LAL6XXX
Department	SoLA	L-T-P [C]	3-0-0
Offered for	MSc	Type	Core for MSc CSS, Elective for others
Prerequisite	Nil		

Objectives:

- To help students conceptualise science and technology as embedded in culture, politics and society, and understand how technical knowledge is produced in society.
- To help students explore the dynamic relationship between technology and society and discover how technology and society influence each other.
- To make students conversant with important and new concepts in science and technology studies in social anthropology.

Learning Outcomes:

The students are expected to have the ability to:

- Comprehend technological progress as a social process
- Find connections between science, technology and culture and think beyond simple binaries like ‘natural’ and ‘social’ sciences and ‘nature’ and culture’
- Develop sensitivity towards an inclusive, sustainable use of technology and appreciate the significance of diversity in science studies

Contents:

Introduction to the Sociology of Science: Introduction to the subject, its founding concepts and traditions, methods of doing STS studies, History of Science and its philosophical traditions, logical positivism, falsification and its critique (7 lectures)

Methodology of Science and its social connections: Structure of scientific revolutions, Ethics and ethos in science and deviance in the practice of science, the Matthew Effect, practising 'agnostic' science, scientific inventions as cultural capital, and the social life of scientists. (8 lectures)

Social Construction of Science*:* Methods of representation in science, Ethnography of a laboratory, social construction of ‘realities’, social construction of nature, Effect of social understanding of race and gender on knowledge (8 lectures)

STS as Social Science: Actor Network theory, Pasteurisation and the world of non-humans, influence of science studies on social sciences, ecological understanding of the world, towards an inclusive understanding of the universe through STS. (8 lectures)

Science and Technology in India: Science & Technology in Colonial India, Development of Indian Science, Science and Governance in India (8 lectures)

Textbooks

- Sismondo, S. (2010), *An Introduction to Science and Technology Studies*, 2nd Edition, Wiley Blackwell
- Miller, D. et. al. edited (2012), *Digital Anthropology,* Berg.

Reference Book

- Felt U., Fouche, R. et. al. edited (2017), *The Handbook of Science and Technology Studies*, MIT Press

Online Course Material

- Mallick, S. *Science, Technology and Society,* NPTEL Course Material, Department of HSS, IIT Guwahati, https://onlinecourses.nptel.ac.in/noc17_hs26/preview
- Mishra, A. *Sociology of Science,* NPTEL Course Material, Department of HSS, Indian Institute of Technology Roorkee, https://onlinecourses.nptel.ac.in/noc18_hs09/preview

Title	**Ethics for Computational Social Sciences**	**Course No.**	LAL6110
Department	SoLA	Structure	3-0-0
Offered for	MSc and PhD	Type	Core for MSc CSS, Elective for others
Prerequisite / co-requisite	None		

Objectives:

- To enable the students to understand and recognise the potent ethical issues involved in social science research and methods
- To inform the students about the ethical concerns in the use of computational tools for data collection and analytics
- To help students develop strategies to identify and address ethical challenges in their professions in computational social science.

Learning Outcomes:

- An ethically sensitive and informed approach to computational tools and Big Data in social science research and analysis.
- An ethically reflective and socially responsive approach to digitised data

Contents:

- What ethics is not, and what ethics is, contemporary relevance of ethical norms and values (10 lectures)
- Ethics in Social Science research and methods, data collection: Informed consent: The process and the ethical concerns; Participant's rights as a morally autonomous agent; Human Rights, data ethics: Data collection, analysis, dissemination: Impact at individual and social level, individual prejudices, entrenched discrimination and data, data ownership and the power dynamics underlying the socio-technical transformation (11 lectures)
- Data protection, anonymisation, and ethical concept of privacy, Big Data: Whose voice is not heard?, social equality, social equity and the digital age (10 lectures)
- Responsible AI and ethical principles, a value-based human approach in Big Data (8 lectures)

Selected references:

1. Uwe Engel, Anabel Quan-Haase, *et al*. (2021). Theory, Case Studies and Ethics. *Handbook of Computational Social Science*. Volume 1. European Association of Methodology Series.
2. Gry Hasselbalch. (2021). *Data Ethics of power*. Edward Elgar Publishing Co.
3. Jeff Collmann & Sorin Adam Matei (eds.). (2016). *Ethical Reasoning in Big Data: An Exploratory Analysis.* Springer.
4. Jaap Bos, Friso Hoeneveld *et al*. (2021). *Research Ethics for Students in Social Sciences*. Springer Nature.
5. M. Coeckelbergh. (2020). AI Ethics (The MIT Press Essential Knowledge Series). The MIT Press.

Title	Language, Society and Computation	Number	LAL 6XXX
Department	School of Liberal Arts	L-T-P [C]	3-0-0 [3]
O↵ered for	UG / PG	Type	Core: MSc CSS
Prerequisite	Nil		Elective: Others

Learning Objectives:

- To explore language in social and computational contexts.
- To understand linguistic diversity with reference to Indian languages.
- To provide a foundation in Natural Language Processing (NLP).

Learning Outcomes:

The students will be able to:

- Understand key concepts in linguistics.
- Analyse multilingual practices in India.
- Demonstrate familiarity with NLP concepts.
- Apply basic NLP tools for text analysis.

Contents:

Overview of Language and Linguistics: Design features of human language; language vs communication systems. Levels of linguistic analysis. Introduction to linguistic data and methods. Spoken vs written language. Basic concepts in phonetics and phonology. (8 lectures)

Morphology: Morphemes, inflexion and derivation, typological variation. Syntax: basic sentence structure and word order variation. Semantics and pragmatics: meaning, context, and interpretation. Illustrations from English and Indian languages. (9 lectures)

Language Families in India: Indo-Aryan, Dravidian, Tibeto-Burman, and Austro-Asiatic. Multilingualism, code-switching, and code-mixing. Grammatical features in Indian languages (agreement, gender, honorificity). Scripts and writing systems. Language, identity, and society. (9 lectures)

Overview of NLP: Scope and applications. Text as data: corpus and preprocessing. Tokenisation, normalisation, and basic text processing. Linguistic annotation: part-of-speech tagging and named entity recognition. (8 lectures)

Implementations and Challenges: AI applications, Indian language challenges, bias, and ethics. (5 lectures)

Text Books and Readings:

- Bird, S., Klein, E., & Loper, E. (2009). *Natural language processing with Python*. O'Reilly Media.
- Fromkin, V., Rodman, R., & Hyams, N. (2018). *An Introduction to Language* (11th ed.). Cengage Learning.

Reference Books:

- Mishra, R. K. (2010). *Language and cognition in multilingual India*. Oxford University Press.
- Sproat, R. (2010). *Language technology for low-resource languages*. Springer.
- Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Online Sources:

Hugging Face NLP Course (Open Access): https://huggingface.co/learn/llm-course/en/chapter1/1

Introduction to NLP (University of Massachusetts – Lecture Materials): https://people.cs.umass.edu/~brenocon/inlp2017/

Course Title	Social Science Research Methods	Course No:	LAL6XXX
Department	SoLA	Structure (L-T-P [C])	3-0-0
Offered for	MSc and PhD	Type	Core for MSc CSS, Elective for others
Prerequisite	Nil		

Objectives

1. To introduce the conceptual foundations of social science research and its major methodological traditions.
2. To develop an understanding of research design, sampling, data collection, and Measurement in social science inquiry.
3. To explain the theoretical basis of probability, statistical inference, hypothesis testing, and model-based reasoning.

Learning Outcomes

1. Students will be able to explain key approaches, concepts, and ethical principles in social science research.
2. Students will be able to frame research problems, formulate research questions, and justify suitable research designs.
3. Students will be able to describe the logic of probability, inference, hypothesis testing, and statistical modelling.

Contents

Foundations and Paradigms of Social Science Research: Meaning, scope, and purpose of social science research; Distinction between social science and natural science research; Major types of social science research; Theoretical, empirical, qualitative, quantitative, mixed-methods, and computational traditions; Research in the digital age; Emerging paradigms in computational social science; Ethical principles, research integrity, and responsible conduct of research. **(6 lectures)**

Research Problem, Theory, and Design: Identification of research problems and research gaps; Formulation of research questions and objectives; Role of theory and conceptual frameworks in research; Review of literature and sources of scholarly evidence; Elements of research design; Concepts, constructs, variables, hypotheses, and levels of Measurement; Qualitative, quantitative, and mixed-methods designs; Cross-sectional, longitudinal, survey, experimental, and correlational designs. (8 lectures)

Qualitative Inquiry, Sampling, and Data Collection: Nature and assumptions of qualitative research; Interpretive and exploratory approaches in social science inquiry; Phenomenology, ethnography, and case study method; Qualitative sampling logic; Credibility, transferability, reflexivity, and researcher positionality; Population, sample, and unit of analysis; Probability and non-probability sampling designs; Sampling bias and sample-size considerations; Sources and types of data; Survey questionnaire, interview, observation, focus group discussion, and participant observation as data collection strategies. (9 lectures)

Measurement, Mathematical Foundations, and Probabilit: Principles of Measurement and scaling; Validity and reliability in Measurement; Sets, relations, and functions as foundations for classification and variable relationships; Vectors and matrices as representations of data; Systems of linear equations and their relevance to statistical modelling; Distance, similarity, and projection as ideas for comparing observations; Basic concepts of optimisation, objective functions, constraints, minimisation, and maximisation; Fundamentals of probability; Random variables; Discrete and continuous probability distributions; Sampling distributions; Estimation, uncertainty, and confidence intervals. (8 lectures)

Statistical Inference and Model-Based Reasoning: Role of statistics in social science explanation; Descriptive and inferential reasoning; Logic of statistical generalisation from sample to population; Hypothesis testing; Null and alternative hypotheses; Significance level, p-value, test statistic, and statistical decision-making; Conceptual basis of z-test, t-test, chi-square test, ANOVA, and MANOVA; Association and correlation; Regression as a model of social relationships; General Linear Model framework; Introductory orientation to Generalised Linear Models for categorical, count, and non-normal outcomes. (8 lectures)

Text Books

1. Ross, S.M. (2014). *Introduction to Probability and Statistics for Engineers and Scientists*, Academic Press.
2. Babbie, E. R. (2020*). The Practice of Social Research*. Cengage Learning.

Reference Books:

1. Seale, C., Gobo, G., Gubrium, J.F. & Silverman, D. (2004). *Qualitative Research Practice*. London: Sage.
Reason, P. (Ed.). (1988). *Human inquiry in action: Developments in new paradigm research*. London: Sage.
2. Ross, S.M. (2014). *Introduction to Probability Models*, Academic Press
3. Kothari, C.R. (2004). *Research Methodology: Methods and Techniques*. New Age International Publishers: New Delhi.

Online Course Material

Kumar, S. Statistical Inference, IIT KGP, NPTEL course material: https://nptel.ac.in/courses/111/105/111105124/
Bajpai, G.S., Research Methodology, NLU Delhi, SWAYAM: https://onlinecourses.swayam2.ac.in/cec20_hs17/preview

Title	Computational Methods for Social Sciences	Number	LAL6XXX
Department	SoLA	L-T-P [C]	3-0-0
Offered for	M.Sc.	Type	Common Core
Prerequisite	Nil		

Objectives:

- Introduce key concepts, principles, and frameworks underlying computational approaches to the study of social phenomena.
- Develop an understanding of the theoretical foundations of data visualisation, agent-based modelling, data extraction, social network analysis, and text mining.
- Enable students to evaluate the scope, relevance, and limitations of computational methods in social science research.

Learning Outcomes:

- Students will be able to explain the principles and theoretical foundations of computational methods used in social science research.
- Students will be able to describe the role of data visualisation, social simulations, network analysis, and text mining in understanding social processes.
- Students will be able to assess the strengths, limitations, and applications of computational techniques in addressing social science questions.

Contents:

Data Visualisation –Data Visualisation Theory, Data Visualisation Through Their Graph Representations, High-dimensional Data Visualisation (7 lectures)
Agent-Based Modelling – Social Simulations and techniques, Introduction to Agent-Based Modelling, Design, Calibration (8 lectures)
Data Extraction:–Introduction to Automated Extraction Information Systems, Tools to Work with Big Data (10 lectures)
Social Network Analysis - Introduction to Social Network Analysis, Types of networks, Measurement of centrality, case studies (7 lectures)
Text Mining - Text mining and extraction, Text preprocessing, Techniques for text analysis (7 lectures)

Text Books:

- Ciofi-Revilla, C. 2014. *Introduction to Computational Social Sciences: Principles and Applications.* Springer: New York

Reference Books:

- Murray, D. 2013. *Tableau Your Data! Fast and Easy Visual Analysis with Tableau Software.* Wiley: London
- Wilensky & Rand. 2015. *An Introduction to Agent-Based Modelling: Modelling Natural, Social, and Engineered Complex Social Systems with NetLogo.* MIT Press: New York
- Kwartler T. 2017. *Text Mining in Practice with R.* Wiley: London
- Chisholm, A. 2013. *Exploring Data with RapidMiner.* Packt Publishing: Amsterdam

Online:

Computational Social Science Specialisation
Instructor: Martin Hilbert, UC Davis
Platform: Coursera
Online link:
https://www.coursera.org/specializations/computational-social-science-ucdavis?

Title	Computational Social Sciences Lab	Number	LAP6XXX
Department	**SoLA**	**L-T-P [C]**	**0-0-2**
Offered for	**M.Sc.**	**Type**	
Prerequisite	**Nil**		

Objectives:

- To develop students' understanding of computational methods for analysing Social Data
- To introduce students to foundational techniques in computational social science through data analytics, programming in Python, and visualisation tools for social data analysis
- To equip students with practical skills in the simulation and modelling of social systems through computational experimentations.

Learning Outcomes:

The students are expected to have the ability to:

- Apply Python-based analytical tools to manage, analyse, and visualise social science data.
- Analyse textual data to understand and interpret social phenomena.
- Design, implement, and evaluate agent-based models to explain social interactions.

Contents:

- Introduction to data analytics and Python fundamentals – Numpy, Pandas, Matplotlib, Seaborn (7 lectures)
- Text analytics - Lexical Semantics, Topic Models, Entity Linking, Information Extraction, Sentiment Analysis (8 lectures)
- Social Network Analysis – NetworkX, Network visualisation and centrality measurements (5 lectures)
- Agent-Based Modelling – NetLogo, Model design, calibration and implementation (6 lectures)

Textbooks:

- John McLevey. (2022). *Doing computational social science: A practical introduction*. SAGE Publications Ltd

Online Materials:

Python For Data Science
Instructor: Prof. Ragunathan Rengasamy, IIT Madras
Platform: NPTEL
Online link: https://nptel.ac.in/courses/106106212

Natural Language Processing
Instructor: Prof. Pawan Goyal, IIT Kharagpur
Platform: NPTEL
Online link: https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs45

Social Network Analysis
Instructor: Prof. Tanmoy Chakraborty, IIIT Delhi
Platform: NPTEL
Online link: https://onlinecourses.nptel.ac.in/e-learning/preview/noc22_cs117

Course Title	Statistics Lab	Course No:	LAP6XXX
Department	SoLA	Structure (L-T-P)	0-0-4 [2]
Offered for	MSc and PhD	Type	Core for MSc CSS, Elective for others
Prerequisite	Nil		

Objectives

1. To build practical foundations in mathematics and statistics for social science research.
2. To train students in using R for data handling, analysis, visualisation, and reporting.
3. To develop applied skills in statistical modelling and computational social science research.

Learning Outcomes

1. Students will be able to use basic mathematical and statistical tools for social science data analysis.
2. Students will be able to work with R, R packages, datasets, and reproducible analytical workflows.
3. Students will be able to apply statistical tests, regression models, and scale validation techniques to social science problems.

Contents

The laboratory component is designed around the following hands-on statistical computing exercises:

Working with sets, relations, and functions using small social science datasets.
Applying set operations for classification, grouping, and filtering of data.
Representing data using vectors, matrices, and simple data tables.
Performing basic matrix operations and solving simple linear equations.
Computing distance, similarity, and projection measures.
Exploring basic optimisation using objective functions and constraints.
Installing and familiarising with R and RStudio.
Creating and managing R objects, vectors, lists, matrices, and data frames.
Installing, loading, and using core R packages for data analysis.
Importing datasets from CSV, Excel, and online sources using R.
Cleaning, transforming, recoding, and preparing datasets using R.
Handling missing values, outliers, and variable transformations using R.
Generating descriptive statistics and visualisations using R.
Simulating probability distributions using R.
Generating sampling distributions and computing confidence intervals using R.
Conducting z-test, t-tests, chi-square tests, and Fisher's exact test using R.
Conducting one-way ANOVA, two-way ANOVA, repeated measures ANOVA, post-hoc tests, and selected MANOVA examples using R.
Performing correlation, linear regression, logistic regression, and Poisson regression using R.
Checking assumptions, computing effect sizes, generating diagnostics, model summaries, and statistical tables using R.
Conducting scale construction and validation using R, including Cronbach's alpha, KMO test, Bartlett's test, PCA, EFA, and introductory CFA.

Text Books

1. Field, A. (2026). *Discovering Statistics Using R and RStudio* (2nd ed.). SAGE.
2. Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for Data Science: Import, Tidy, Transform, Visualise, and Model Data* (2nd ed.). O'Reilly.

Reference Books

Ross, S. M. (2014). *Introduction to Probability and Statistics for Engineers and Scientists*. Academic Press.

Online Course Material
Stanford Online: Statistical Learning with R
https://online.stanford.edu/courses/sohs-ystatslearning-statistical-learning-r?utm_

Second Semester Compulsory Courses

Title	Computational Methods in Indian Knowledge Systems	Number	LAL6XXX
Department	School of Liberal Arts	L-T-P [C]	3-0-0 [3]
Offered for	MSc	Type	Core for MSc and Elective for Others
Prerequisite	None		

Objectives

1. To introduce students to the computational and algorithmic foundations embedded in classical Indian knowledge traditions.
2. To explore mathematical, linguistic, and astronomical texts as sources of formal reasoning and structured computation.
3. To understand combinatorial, recursive, and algorithmic thinking in texts related to prosody (*Chandas*), astronomy (*Jyotiṣa*), and mathematics (*Gaṇita*), while encouraging interdisciplinary research that connects Indian Knowledge Systems (IKS) with contemporary scientific and technological frameworks.

Learning Outcomes

The students will be able to

1. Identify and explain computational principles inherent in classical Indian texts.
2. Analyse traditional rule-based systems and map them to modern computational models.
3. Critically evaluate the relevance of Indian Knowledge Systems in modern scientific discourse and emerging technologies, while designing interdisciplinary projects that meaningfully integrate traditional knowledge with contemporary scientific and technological frameworks.

Contents:

1. **Introduction to IKS and Computation:** Overview of Indian Knowledge Systems, Computational thinking: concepts and relevance, Algorithms, formal systems, and knowledge encoding **(5 Lectures)**
2. **Knowledge Representation in *Śāstra* Tradition:** *Sūtra* methodology: compression and structure, Commentarial traditions, Ontology and categorisation **(5 Lectures)**
3. **Computational Linguistics – Pāṇinian System:** Structure of *Aṣṭādhyāyī*, *Māheśvara-sūtra* and encoding, Transformations and recursion **(5 Lectures)**
4. **Combinatorics in *Chandas*:** Piṅgala's system of prosody, Binary representation and recursion, *Meru-prastāra* (Pascal triangle equivalent), Algorithmic generation of metres **(5 Lectures)**
5. **Computational Astronomy:** *Siddhāntic* traditions and models, Mean vs true planetary positions, Time computation and calendrical systems, Eclipse computation techniques **(9 Lectures)**
6. **Indian Mathematics and Algorithms:** Indian mathematicians and their major contributions. Arithmetic, geometry, algebra, series and recursive techniques discussed in ancient Indian Sanskrit mathematical texts **(10 Lectures)**

Text Books:

1. *Āryabhaṭīya* of Āryabhaṭa, ed. with tr. and notes by K. S. Shukla and K. V. Sarma, INSA, New Delhi 1976.
2. *History of Hindu Mathematics*, B. B. Datta and A. N. Singh, 2 vols, Lahore 1935, 1938 (repr. Asia Publishing House, Bombay 1962; repr. Bharatiya Kala Prakashan, Delhi 2004).

Reference Books:

3. G. Emch, M. D. Srinivas and R Sridharan (eds), Contribution to the History of Mathematics in India, Hindustan Publishing Agency, New Delhi, 2005.
4. Seshadri C. S. (2010). Studies in the History of Indian Mathematics, Hindustan Book Agency. New Delhi 2010.

Online Course Materials

1. IKS Division, Ministry of Education, Lectures and Videos, https://iksindia.org/lectures-and-videos.php

Course Title	Health and Society	Course No.	LAL6XXX
Department	SoLA	Structure (L-T-P-C)	3-0-0 [3]
Offered for	MSc CSS	Type	Core for MSc CSS and Open elective for others
Prerequisite		None	

Objectives

- To understand the inter-relationship between social determinants, economic conditions, and cultural norms on health
- To understand health systems in the current societal context of India, in a comparative context with the rest of the world.
- To learn the usage of various health data in research, policy making, and governance.

Learning Outcomes

The students will be able to

- Learn the role of social determinants on health.
- Understand various aspects of the Indian health systems and cross-country comparison.
- Explore and learn about the use of various secondary data and their implications for research, policy-making, and governance.

Contents

Measurement of health: Understanding the basic concepts of holistic health and healthcare; Healthcare transitions in India and the world; Emergence of non-communicable diseases. (7 lectures)

Social determinants of health: Role of socio-economic factors on health, Gender and health, culture and health, health inequalities and inequities, social isolation and mental health. (7 lectures)

Context, crisis, and health: environmental crisis; workplace practices and burnout; humanitarian crises and conflicts; technology and the mental health pandemic. (5 lectures)

Health systems: Understanding health systems, health system frameworks, community mental health, system thinking approaches, service delivery, health workforce, health technology, healthcare financing, leadership, and governance. (5 lectures)

Contemporary issues in health care: mental health, child and maternal health, elderly health, disability, and palliative care; access to healthcare: treatment gap, stigma, and awareness; burden of disease; and digital health. (5 lectures)

Data sources in health research: National Sample Survey; Family Health Survey; Longitudinal Ageing Studies in India; Study on global AGEing and adult health (SAGE); National Mental Health survey; Mental Health Atlas (9 lectures)

Textbook

· Detels, R., Beaglehole, R., Lansang, M. A., & Gulliford, M. (2011). Oxford textbook of public health. Oxford University Press.

Reference Books/Articles

· Abookire, S., Plover, C., Frasso, R., & Ku, B. (2020). Health design thinking: an innovative approach in public health to defining problems and finding solutions. Frontiers in public health, 8, 459.

· Rosen, G. (2015). A history of public health. Jhu Press.

· Birn, A. E., Pillay, Y., & Holtz, T. H. (2017). Textbook of global health. Oxford University Press.

· Perianayagam, A., Bloom, D., Lee, J., Parasuraman, S., Sekher, T. V., Mohanty, S. K., ... & Pramanik, S. (2022). Cohort profile: the longitudinal ageing study in India (LASI). International Journal of Epidemiology, 51(4), e167-e176.

Online Reading Materials

- Health and Society (University of Chicago)
  http://collegecatalog.uchicago.edu/thecollege/healthandsociety/

Course Title	Introduction to Policy Studies	Course No.	LAL6130
Department	SoLA	Structure (L-T-P-C)	3-0-0 [3]
Offered for	MSc CSS	Type	Core for MSc CSS and Open elective for others
Prerequisite	None		

Objectives:

- To introduce Policy Studies to students
- To understand how policy is important in a digital world
- To introduce digital tools for policy analysis

Learning Outcomes:

The students are expected to have the ability to:

- An awareness of policy studies and policy analysis
- To be introduced to policy in digital era
- Knowledge to work in the field of Policy analysis

Contents:

I**ntroduction**: Concept and Theories, what is Policy, Public Policy, Practical Analysis of Public Policy, Policy Types: Distribution, Regulation, Redistribution, (11 lectures)

Understand relationship between State and Policy - The making of the Therapeutic State, Policy growth, Redistribution of Public goods as policy (11 lectures)

Morality, Law, and Policy: Morality and Policy, Law and Policy, Politics and Policy (9)

Policy in current time: Introduction to Fiscal Policy, Policy in a Digital World (8)

Text Books:

- Knill, C. 2012. Public Policy: A New Introduction. Palgrave Macmillan: New York

Reference Books:

- Kagami et. al 2004. Information Technology Policy and the Digital Divide. Springer: London
- Andreasson, K. 2015. Digital Divides: The New Challenge and Opportunities of e-Inclusion. Taylor & Francis: London

Online Resources

https://www.isb.edu/en/study-isb/advanced-management-programmes/amp/public-policy.html

Course Title	Social Realities	Course No.	LAL6070
Department	SoLA	Structure (L-T-P-C)	3-0-0 [3]
Offered for	MSc CSS	Type	Core for MSc CSS and Open elective for others
Prerequisite	None		

Objectives:

- To examine hierarchical differentiation that characterises human society
- To analyse ideological and practical explanations offered to justify social inequalities
- To identify the intersectionalities that create and sustain social inequalities

Learning Outcomes:

The students are expected to have the ability to:

- Develop sensitivity to social inequality
- Develop reflexivity and understand privilege
- Address issues of social inequality while designing technical solutions

Contents:

Introduction: Concept and Theories Introduce the concept of social stratification and most significant theories around the concept (3 lectures)

Forms of Stratification: Class, Caste and Race - Theories and empirical studies on class analysis and the transformation of class structures; theories on caste; empirical studies with a focus on contemporary India. Caste in Labour markets. Export of caste, religion, Ethnicity (9 lectures)

Class and Caste in India: An overview of the Indian condition (8 lectures)

Stratification and Gender: Gender inequalities, labour market and inequalities, intersectionality of Gender with class, caste and race (6 lectures)

Race, Ethnicity and stratification: Race and social inequality, Race and colonisation (8 lectures)

Gender in a global state: Gender and race intersectionality (5 lectures)

Text Books:

A handbook with all the study material will be given to the students.

Reference Books:

- Gupta, D. ed. 1991. Social Stratification. Delhi: Oxford University Press.
- Tumin, M. M. 1953. Some Principles of Stratification: A Critical Analysis. American Sociological Review, 18 (4): 387-94
- Jodhka, S. S. 2012. Caste. Oxford India Short Introductions. Delhi: Oxford University Press
- Acker, J. 1998. Women and Social Stratification: A Case of Intellectual Sexism, in Feminist Foundations: Towards Transforming Sociology, Kristen et. al. (ed.) Delhi: Sage.
- M.N. Srinivas. 1996. Caste: Its 20th Century Avatar. Penguin.

Online Source:

NPTEL course https://onlinecourses.nptel.ac.in/noc20_hs86

Course Title	Applied Economic Foundation for Social Sciences	Course No.	LAL6XXX
Department/School	SoLA	Structure (L-T-P)	3-0-0
Offered for	Msc CSS	Type	
Prerequisite		None	

Objectives

- Provide students with a foundational understanding of microeconomic theory and its applications.
- To develop an understanding of key macroeconomic concepts, models, and policy tools.
- Introduce students to strategic thinking and decision-making in economic contexts.

Learning Outcomes

- By the end of the course, students will be able to:
- Explain and analyse how individuals and firms make economic decisions and how these decisions interact in various market settings.
- Construct and critically evaluate macroeconomic models, such as the IS-LM model; explain the mechanisms of monetary and fiscal policy; and assess open-economy dynamics and growth theories.
- To analyse the complex interrelationships between economic growth, development, and institutions

Contents

Module 1: Microeconomics: Preferences, utility functions, and choice; Budget constraints, Elasticities and optimal choice; Production theory – isoquants, cost minimisation; Profit maximisation and cost functions; Perfect competition and partial equilibrium; Monopoly and market power; Analysis of Demand and Supply; Game theory basics – Nash equilibrium; Externalities and public goods; Asymmetric information – adverse selection, moral hazard; Case studies and Models (13 hours)

Module 2: Macroeconomics: Classical model – labour market, production, saving/investment

IS-LM model – goods and money markets; Theories of Consumption and Investment; Money demand; Money supply, and interest rates; Inflation and Phillips Curve; Fiscal policy; Monetary policy; Business cycle facts and stylised models; Open economy basics – exchange rates, trade balance; Case studies and Models (13 hours)

Module 3: Development Economics: Growth and development; Capabilities approach and human development frameworks; Role of state in development; Institutional economics and development; Concepts and measures of poverty; Labour market transitions; Information Economics; Digital economy and tech disruptions; Sustainable development frameworks; Case studies and Models (13 hours)

Text Books:

- Samuelson, P.A. & Nordhaus, W.D. (2020). Economics (23rd ed.). McGraw-Hill.
- Mankiw, N. G. (2021). *Principles of economics* (9th ed.). Cengage Learning

Reference Books

- Ray, D. (1998). Development economics. Princeton University Press.
- William Branson (2022). Macroeconomic Theory and Policy, 3rd Edition, East West Press
- Varian Hal R. (2020). Intermediate microeconomics: a modern approach. Springer

Online resource:
StanfordOnline: Principles of Economics
https://www.edx.org/learn/economics/stanford-university-principles-of-economics

Title	**AI & ML for Social Sciences & Humanities**	Number	LAL6XXX
Department	SoLA	Structure L-T-P	3-0-0 (3 credits)
Offered for	M. Sc. CSS, M. Sc. DH	Type	Compulsory
Prerequisite	Statistics Lab		

Objectives:
1. To make social science and humanities students conversant in AI & ML concepts and basic programming
2. To introduce AI-based applications that will help students in research and jobs
3. To introduce students to critical perspectives on AI

Learning Outcomes:
1. Demonstrate knowledge of AI & ML concepts and programming for use in Social Sciences and Humanities.
2. Use particular AI applications for professional purposes
3. Evaluate AI critically and demonstrate knowledge on the ethical use of AI

Contents:
Knowing the AI ecosystem: What is intelligence?; Defining Artificial Intelligence; Evolution of AI, Importance of AI today; Different areas of AI and where social scientists can use it; AI & Policy: India AI mission; *Case Study* (5 lectures)
Machine Learning concepts and types: Theory behind machine learning and understanding terminologies; From Data to Model: Exploring approaches to ML; Mathematics and CS in Machine Learning; Supervised learning and its working; Unsupervised learning and reinforcement learning; *Case Studies: Real-world applications (13 lectures)*
Deep Learning and Generative AI: An introduction to deep learning; What are LLMs?; Understanding Generative AI; Prompt engineering for social science and humanities tasks; AI for
academic reading and writing; AI in qualitative and quantitative research, Case Study (10 lectures)
Critical Perspectives on AI: The narrative construction of AI; Data and AI; Bias and AI; Emotions and AI; Belongingness and AI (13 lectures)

Text Books:
- Soliman, K. S., 2021, *Artificial Intelligence and Machine Learning,* Springer, London
- Guttag, John V., 2021, *Introduction to Computation and Programming using Python,* MIT Press: Boston

References:

- Liu, T., Webb, J. et al., 2024, *AI2023: Advances in Artificial Intelligence,* Springer: New York
- Theobald, O., *Machine Learning for Absolute Beginners: A Plain English Introduction,* Booky Publishers

Online resources:

An introduction to Artificial Intelligence, IIT Delhi (Prof. Mausam). https://nptel.ac.in/courses/106102220

Google AI essentials. https://grow.google/ai-essentials/

Title	**AI & ML for Social Sciences & Humanities-Lab**	Number	LAP6XXX
Department	SoLA	Structure L-T-P	0-0-2 (1 credit)
Offered for	M. Sc. CSS, M. Sc. DH	Type	Compulsory
Prerequisite	None		

Objectives:

1. To develop programming and logic skills in students
2. To be conversant in Python programming and use it for different applications

Learning Outcomes:

1. Develop programming logic and think in terms of flowcharts and algorithms
2. Demonstrate skills and knowledge in Python programming

Contents:

The lab component includes the following hands-on exercises:

- Installing and running Python (1)
- Understanding variables and data types (2)
- User input and basic operations (3)
- Conditional statements (2)
- Loops, lists and functions (2)
- Working with files (2)
- Error handling and debugging (2)
- Data analysis with Python (introduction to pandas); (6)
- Python and automation for researchers - SciKit Learn (6)

Text Book:

Guttag, John V., 2021, *Introduction to Computation and Programming using Python,* MIT Press: Boston

Online resources:

https://www.youtube.com/watch?v=eoPsX7MKfe8&list=PLIdgECt554OVFKXRpo_kuI0XpUQKk0ycO

(IIT Kanpur NPTEL program for Python programming)

PROGRAMME ELECTIVE

Course Title	**Contemporary Western Philosophy**	Course No.	LAL7940
Department/School	SoLA	Structure (L-T-P-D)	3-0-0 [3]
Offered for	PhD	Type	Elective
Prerequisite		None	

Objectives

- Appreciate major developments and figures in contemporary Western philosophy
- Analyse philosophical arguments from different schools and traditions
- Evaluate analytic and continental approaches to philosophical problems

Learning Outcomes

- Identify and explain the key ideas of major contemporary philosophers
- Critically assess arguments from schools such as existentialism, analytic philosophy, and post-structuralism
- Distinguish between different philosophical methodologies and frameworks

Contents

- Origins: Overview of Contemporary Western Philosophy, Major Characteristics, Rationalism, Empiricism, Idealism (4 lectures)
- Phenomenology and Existentialism: Husserl, Merleau-Ponty, Heidegger, Sartre, de Beauvoir (6 lectures)
- Analytic Philosophy: Frege, Russell, Wittgenstein, Quine, Davidson, Moore (8 lectures)
- Moral Philosophy: Nagel, Singer, MacIntyre (6 lectures)
- Political Philosophy: Nozick, Rawls, Habermas, Pogge, Sen (5 lectures)
- Philosophy of Mind: Dennett, Chalmers, Fodor, Searle, Churchland (5 lectures)
- Postmodernism: Lyotard, Foucault, Derrida (5 lectures)

Text Book

- John G. Cottingham(2021). Western Philosophy: An Anthology. Blackwell Philosophy Anthologies
- *Stephen P. Schwartz(2012)A Brief History of Analytic Philosophy Wiley Blackwell 2012*

Reference Books

- Dermot Moran ed (2008)The Routledge Companion to Twentieth Century Philosophy.Routledge
- *Gutting, Gary. (2001)French Philosophy in the Twentieth Century*

Online Reading Materials

Aspects of Western Philosophy NPTEL course: Humanities and Social Sciences - Aspects of Western Philosophy

Course Title	Human Behaviour in Context: Neuro-cultural and Digital Perspectives	Course No.	LAL 7930
Department/School	SoLA	Structure (L-T-P-C)	3-0-0 [3]
Offered for	PhD	Type	elective
Prerequisite		None	

Objectives

The Instructor will

1. Explain foundational and contemporary theories explaining human behaviour.
2. Enable students to analyse complex behavioural phenomena using interdisciplinary theoretical frameworks.
3. Provide empirical evidence from diverse methodologies and perspectives in behavioural science.

Learning Outcomes

By the end of this course, students should be able to

1. Identify major human behaviour theories and models (e.g., dual-process theory, self-determination theory, predictive processing).
2. Apply theoretical models to analyse contemporary behavioural challenges.
3. Judge the effectiveness of intervention models for habit formation, identity change, or behavioural resilience.

Lecture Modules/Contents

Theoretical Foundations and Meta-Theoretical Perspectives: Defining Human Behaviour: Scope, Complexity, and Context; Meta-Theoretical Foundations; Classical Theories; The Biopsychosocial Model and Integrative Frameworks; Evolutionary Psychology and Cultural Evolution in Behaviour (9 lectures)

Biological and Neurological Mechanisms: Reward Systems, Reinforcement, and Dopaminergic Circuits; Emotional Regulation Systems; Neuroplasticity and Epigenetic Adaptation; Interaction of Emotion and Cognition; Cognitive Dissonance, and Belief Rigidity (8 lectures)

Motivation, Goal Systems, and Behavioural Regulation: Multi-System Models of Motivation; Value-Based Decision-Making and Neuroeconomics; Habits, Automaticity, and Context-Dependent Behaviours; Digital Distraction and the Attention Economy (6 lectures)

Social, Cultural, and Power Dynamics: Social Influence; Identity Complexity and Intersectionality; Power, Status, and Social Dominance Theories; Collective Behaviour and Social Contagion; Behaviour in Digital Networks and Algorithmic Systems; (7 lectures)

Behavioural Change, Intervention, and Emerging Frontiers: Change Models: COM-B, Nudges, and Fogg's Framework; Behaviour Change at Scale; Breaking and Building Habits in Context; Predictive Behavioural Analytics and Digital Phenotyping; Global and Cultural Dimensions of Behavioural Science (9 lectures)

Text Book

1. Namy, L., Woolf, N., Lilienfield, S., & Lynn, S. (2014). Psychology: From Inquiry to Understanding. Pearson Education Limited, Global 3rd Edition, ISBN: 1292058846.

Reference Books

1. Uslaner, E.M. (2018). The Oxford Handbook of Social and Political Trust. Oxford University Press. ISBN: 9780190274801
2. Hohwy, J. (2013). The Predictive Mind. Oxford University Press. ISBN: 9780199682737
3. Dihigg, C. (2014). The Power of Habit: Why We Do What We Do in Life and Business. Random House. ISBN: 081298160X

Online resources

1. Richardson, G.D. Advanced Psychology: A Comprehensive Guide to Human Behaviour. https://www.udemy.com/course/midday-psychology/?couponCode=ST8MT220425G3. Udemy

Title	Cognition & Emotion	Number	LAL 7910
Department	School of Liberal Arts	L-T-P [C]	3-0-0 [3]
Offered for	PG and PhD	Type	Elective
Prerequisite	Nil		

Learning Objectives:

1. To provide an understanding of human Cognition within individual and social contexts.
2. To equip students with a theoretical foundation of emotion and its interaction with Cognition.
3. To develop an understanding of variability in cognitive-affective processes across individuals and cultures.

Learning Outcomes:

The students will be able to:

1. Understand key concepts and theoretical models in cognitive and affective psychology.
2. Explain the processes of attention, memory, and social inference in human Cognition.
3. Critically analyse the interaction between emotion and cognitive processes, including decision-making and executive functioning.
4. Integrate multiple levels of analysis—cognitive, psychological, and neurological—to understand human behaviour.
5. Examine variability in Cognition and emotion across individual, social, and cultural dimensions.

Course Modules:

1. *Module 1: Foundations of Cognition and Social Cognition*
   Overview of Cognition: Historical and modern views. Philosophical approaches: Rationalism, empiricism, constructivism, and embodied Cognition. Key theories in cognitive science (information processing, connectionism, situated Cognition) with relevance to emotion. Introduction to social Cognition as emotionally grounded. Dual-process models (System 1 & 2) and their emotional underpinnings. The social brain, emotional awareness, and self-regulation. (10 lectures)

2. *Module 2:Inferences, Judgments, and Biases in Social Cognition*
   Attribution processes and emotional influences. Heuristics and affective biases in judgment. Emotion in counterfactual thinking and decision-making. Affect vs accuracy in social judgments. Emotional roots of stereotyping and implicit bias. Culture, language, and emotionally-driven social Cognition. Attitudes as affective-cognitive structures. (10 Lectures)

3. *Module 3:Emotion and Cognition*
   Core theories of emotion (psychological and cognitive). Emotion's role in attention, memory, decision-making, and consciousness. Executive function and emotional regulation. Emotion-cognition conflict in stress, procrastination, and coping. (11 lectures)

4. *Module 4: Variability in Emotion and Cognition*
   Gender and emotional expression and processing. Emotional and cognitive changes with ageing. Individual differences in emotional reactivity and regulation. Cultural and social shaping of emotional experiences. Affective disorders: Neural and cognitive dimensions. (8 lectures)

Textbooks and readings:

- Lemaire, P. (2021). "Emotion and cognition: An introduction". Routledge.
- Fiske, S. T., & Taylor, S. E. (2020). "Social cognition: From brains to culture" (3rd ed.). SAGE Publications Ltd.
- Pessoa, L. (2013). "The cognitive-emotional brain: From interactions to integration". MIT Press.

Reference Books:

- Adolphs, R., & Anderson, D. (2018). *The neuroscience of emotion: A new synthesis*. Princeton University Press.
- Srinivasan, N. (2019). *Emotion and cognition* (1st ed.). Academic Press.
- Fiske, S. T., & Macrae, C. N. (Eds.). (2012). *The SAGE handbook of social Cognition*. Sage.
- Pessoa, L. (2013). *The cognitive-emotional brain: From interactions to integration*. MIT Press.

Online Sources:
https://nptel.ac.in/courses/109/106/109106150/
https://www.coursera.org/learn/emotions
https://study.sagepub.com/fiskeandtaylor3e

Course Title	**Historical and Social Anthropology**	Course No.	LAL7890
Department	School of Liberal Arts	L-T-P[C]	3-0-0 [3]
Offered for	Ph.D. coursework	Type	Elective
Pre-requisite	None		
Objectives			

1. Introduce and explain important texts and methods in social, historical, and anthropological research.
2. Understand and engage with the intersections of Sociology, Anthropology and History.
3. Help students apply interdisciplinary research tools in their research.

Learning Outcomes

1. Analyse and integrate historical contexts within sociological and anthropological frameworks to formulate nuanced research questions.
2. Understand the difference between primary and secondary texts.
3. Understand sociological and historical construction of social and state power.
4. Analyse and apply anthropological and historical texts for their own work

Contents

The Basics of Socio-historical Anthropology (10 lectures)

Introduction to early defining works in social and historical anthropology, the relationship between social and historical anthropology, Importance of historical context in sociological research, Methods of socio-historical anthropology

Village Studies and Social Identities (7 lectures)

Intersections between village studies and socio-cultural identities such as Gender, caste and class.

Everydayness of the State (7 lectures)

Power and limitation of the state, The state and the margin, Bureaucracy and the state, Ethnography of the state

Historical Sources and Archival Ethnography (7 lectures)

Learning the historians' craft, Understanding the archive, understanding archival ethnography, Ethics of working with an archive

Contemporary Trends and Texts in Ethnography (8 lectures)

The Anthropology of Ontologies, Risky ethnography, Ethnographies of Violence, Ethnography of Space and Place, Anthropology of Work and Labour

Text Books

1. Das, Veena (ed.) 2003 *The Oxford India Companion to Sociology and Social Anthropology*. New Delhi: Oxford University Press.

References

1. Srivastava, S., Arif, Y. and Abraham, J. (eds.) (2019) *Critical themes in Indian sociology*. New Delhi: Sage Publications.
2. Das, V., Poole, D., Das, V., & Poole, D. (eds.) (2004) Anthropology *in the Margins of the State*. New Mexico: School of American Research Press.
3. Bennett, J. W. (2017). *Classic Anthropology: Critical Essays, 1944-96*. New York: Routledge.

Online resources

https://onlinecourses.nptel.ac.in/noc25_de06/preview

Title	**Politics of Water Governance**	**Number**	**LAL7420**
Department	SoLA	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., M.Sc., PhD.	Type	Elective
Prerequisite	Interest in Technology, Policy, Governance, Sustainability, Law, Gender, Politics, Social		

	Justice/Rights, Ecology **or** an interdisciplinary/cross-disciplinary background		

The Instructor will:

- Provide a conceptual sociological critical understanding of water governance through global, national policies and geopolitics, and the multiple challenges societies face in addressing water issues.
- Introduce students/young engineers to examine technologies through the lens of sustainability questions and to understand how technologies impact the environment, industries, and local communities' livelihoods through the case of water.
- Facilitate students' understanding of transdisciplinary pathways to explore conceptual and empirical questions concerning water scarcity, technologies, water pollution, uncertainty/rights and access to water resources, and the cultural politics of water development.

Learning Outcomes:

The students are expected to have the ability to:

- Interpret the theories and concepts underlying multiple levels of water governance and contemporary global water debates around SDGs from a local/regional perspective.
- Reflect on the concepts of sustainability in technology development and analyse the water nexus with livelihoods, technologies, and pollution.
- By the end of the course, students/young engineers will be able to understand the politics and the socioeconomics of water pollution, and the sustainability question of the technologies and their relation to water.

Contents:

1. *Water management across space and time*: historical narrative (4)
2. *Major paradigm shifts in water governance:* groundwater governance; politics and socioeconomics of water pollution, political ecology of water (6)
3. *Water as wicked problem:* water laws/regulations/policy in India, Water conflicts in India and South-Asia trans-boundary water conflicts (5)
4. *Gender and water:* engineering masculinities in water governance (5)
5. *Water rights and water justice:* global politics of water grabbing; new spaces for water justice (5)
6. *Social scientific understanding of socio-technical systems:* politics of dams; virtual water flow, cost-benefit analysis of technologies and the livelihoods surrounding water (6)
7. *Foucauldian discourse analyses:* new water standards and practices of multinational corporations (3)
8. *Water-climate change:* water-food security nexus, water-agriculture nexus, water-energy nexus (5)
9. *Water-centric SDGs:* SDGs 2, 6, 14, India and UN (3)

Reference Books

1. Mehta, L. (2005). *The Politics and Poetics of Water: The Naturalisation of Scarcity of Western India*, Orient Longman.
2. Shah, T. (2009). *Taming the Anarchy: Groundwater Governance in South Asia. Resource for Future,* Johns Hopkins University Press.
3. Zwarteveen. M.Z., Ahmed, S. and Gautam, S.R. (2012). *Diverting the flow: Gender equity and water in South Asia*, Zubaan
4. Sultana, F., and Loftus, A. (ed.) (2012), The Right to Water: Politics, Governance and Social Struggles, Routledge.
5. Lankford, B., Bakker, K., Zeitoun, M., and Conway, D. (eds.) (2013), Water Security Principles, Perspectives and Practices, 1st Edition. Routledge.
6. Naz, F. (2014), The Socio-cultural Context of Water: Study of a Gujarat Village, Orient Blackswan.

7. Shah, M., and Vijayshankar, P.S. (2016). Water: Growing Understanding, Emerging Perspectives (Readings on the Economy, Polity and Society), Orient BlackSwan.
8. Zwarteveen, M., Kemerink-Seyoum, J. S., Kooy, M., Evers, J., Guerrero, T. A., Batubara, B., Wesselink, A. (2017). Engaging with the politics of water governance: Wires Water, 4(6).
9. Boelens, R., and Perreault, Vos, J. (2018). Water Justice, Cambridge University Press

Online Resources

Mukherjee, J., Water, Society and Sustainability. Department of Humanities & Social Sciences, Indian Institute of Technology, Kharagpur, https://nptel.ac.in/courses/109/105/109105136/

Course Title	**Sociology of Education**	Course No.	LAL7040
Department	SoLA	Structure (L-T-P [C])	3-0-0 [3]
Offered for	PhD	Type	Elective

Prerequisite	None		

Course Objectives

The Instructor will:

1. Enable students to understand the different dimensions of Education and its dialectical relationship with the production/ transmission/ transformation of various social structures, categories and identities
2. Expose students to diverse theoretical perspectives and studies that will enable them to critically examine the Indian Education system, its features and actors from a sociological standpoint

Learning Outcomes

After this course, students will:

1. Develop analytical and reflexive skills in reading original texts by educational thinkers
2. Understand the social history and context of seminal changes in Indian Education
3. Cultivate a 'sociological imagination' through identifying points of connection between one's personal milieu and the roles played by educational ideologies and practices in shaping, sustaining and transforming them.

Contents

I. Sociological Frameworks and the main stakeholders in Education (10 Lectures): The individual, social and political dimensions of Education (Theoretical perspectives relating to socialisation, structure, agency and resistance); The political economy of Education (Education as a public and private good; Educational Development, State and Market; Aspirations, work and social mobility; Globalisation and privatisation of Education)

II. Socio-historical Transformations of Indian Education (8 Lectures) Education in Pre-colonial and colonial India; Education Policies and their formation post-Independence (NEP 1966, 1986, 2020); New technologies and Educational Transformations post the 1990s

III. Knowledge, Pedagogy and Curriculum (12 Lectures): Theoretical Perspectives (Social Constructivism; Knowledge and Power; Social Interactionism); Constructions of school knowledge, curricula and textbooks; Teachers: Professional identity and agency; Changing milieus, modes and technologies of learning

IV. Identities and Inequalities (11 lectures): Theoretical perspectives (Social Stratification and Social Reproduction); Mapping Learner Diversities and inequalities; Digital Divide and Learning

Text Book

1. Apple, M., Ball, S., & Gandin, L. (2015). *The Routledge international handbook of the sociology of Education*. London: Routledge

References

1. Barr, R., & Dreeben, R. (2008). How schools work. *Schools and society: A sociological approach to Education*

2. Connell, R. W., & Connell, R. (1993). *Schools & social justice*. James Lorimer & Company. 3. Jeffrey, C. 2010. *Timepass: Youth, Class and the Politics of Waiting in India.* Stanford, California: Stanford University Press

4. Kumar, K., eds. (2017). *Routledge Handbook of Education in India*. Taylor & Francis

Online Resources:

1. Cope, Willian. Learning, Knowledge and Human Development: https://www.coursera.org/learn/learning-knowledge-human-development
2. Alves, Innes. Diversity and Inclusion in Education: https://www.coursera.org/learn/diversity-and inclusion-education

Course Title	**Sentence Structure**	Course No.	LAL7030
Department	School of Liberal Arts	Structure (LTPC)	3-0-0
Offered for	UG, PG and PhD	Type	
Prerequisite	None		

Objectives

- To equip students with fundamental concepts of public health, enabling them to organise their thinking and learning in public health.
- To introduce students to various research methods used in public health.
- To understand the organisation of the health system in different countries and societies to meet their health needs.
- To develop students' critical skills to learn about key public health issues in India and around the world.

Learning Outcomes

By the end of the course, students will be able to:

- Learn fundamental concepts related to public health.
- Explore various research methodologies used in the field of public health.
- Understand various health systems in the world.
- Critically identify various issues in public health.

Contents

- **Foundation of modern public health:** The emergence of public health and the development of discipline, data on populations and mortality, understanding the burden of disease, Inequalities in health, the impact of health care on population health, and assessing the impact on population health of policies in other sectors. (9 lectures)
- **Major determinants of health:** The changing nature of infectious disease**,** recognising contemporary determinants of public health: human rights, armed conflict and genetics, food, trade and health**,** Sustainable development and climate change. (6 lectures)
- **Public health policies, laws, and ethics:** Ethical principles and ethical issues in public health, law and public health, priority setting, social values and public health, leadership in public health, international efforts to promote public health. (7 lectures)

- **Epidemiological and social science techniques:** Epidemiology, cross-sectional studies, principles of outbreak investigation, case-control studies, cohort studies, public health surveillance, demography and public health, health promotion, health education and the public's health. (8 lectures)
- **Major public health problems and needs:** cardiovascular diseases; mental health; child health; geriatric health; obesity; physical activity and health; cancer and public health; musculoskeletal disorders; neurological disorders; and tuberculosis. (9 lectures)

Textbooks

- Detels, R. (Ed.). (2015). *Oxford textbook of global public health*. Oxford University Press, USA.
- Sim, F., & McKee, M. (2011). *Issues in public health*. McGraw-Hill Education (UK).

Reference Books

- Shultz, J. M., Sullivan, L. M., & Galea, S. (2019). *Public health: An introduction to the science and practice of population health*. Springer Publishing Company.
- Gurol-Urganci, I., Campbell, F., & Black, N. (2017). *Understanding health services*. McGraw-Hill Education (UK).
- Birn, A. E., Pillay, Y., & Holtz, T. H. (2017). Textbook of global health. Oxford University Press.

Online Reading Materials

- https://www.coursera.org/learn/foundations-public-health-approach

https://onlinecourses.nptel.ac.in/noc20_hs20/preview

Title	**Theoretical Foundations of Econometrics**	Number	7XXX
Department/School	AIDE (CMCE)	L-T-P [C]	3-1-0 [3]
Offered for	B. Tech., Post-grad, PhD.	Type	Elective
Prerequisite	Statistics		

Objectives:

1. To introduce students to fundamental concepts in econometric theory.
2. To provide knowledge of the theoretical underpinnings of applied econometric methods.
3. To provide an understanding of estimation in econometrics and its associated concepts.

Learning outcomes:

1. Students will gain foundational knowledge of inference and estimation techniques.
2. Students will develop an intuition for the generalizability of different econometric models.
3. Students will understand the links among the theory of econometric models, the computations required for inference, and the application of the models to real-world examples.

Course content:

Revision of key statistical concepts: Descriptive statistics, Random variables, Joint distributions, conditional distribution and their properties, Moment generating functions, Normal distribution and other related distributions, Central Limit Theorem, Population, parameters and random sampling, sampling distributions, chi-square distribution, t distribution, F-distribution (6 lectures)

Estimation: Finite sample properties of estimators – unbiasedness and efficiency, Asymptotic or large sample properties of estimators – consistency, asymptotic normality (4 lectures)

General Approaches to parameter estimation: Method of moments, maximum likelihood estimation method, Least Squares method. Interval estimation and confidence intervals (3 lectures)

Linear Regression Model: multiple linear regression models, Gauss-Markov Theorem, inference and hypothesis testing in the context of linear regression, multiple hypothesis testing, calculating marginal effects, mechanics and interpretation of ordinary least squares, Frisch-Waugh-Lovell Theorem, different econometric tests (LM test, LR test and Wald test) and their principles. (15 lectures)

Issues related to OLS: Heteroskedasticity and White test, Breusch-Pagan/Godfrey Test, Goldfeld-Quandt Test, Generalised Least Squares, Multicollinearity, Functional form misspecification and Ramsey RESET test, specification errors – errors of omission and errors of commission, restricted least squares, omitted variable bias, measurement errors and properties of OLS under measurement error, missing data, outliers and influential observations, sample-selection issues, Endogeneity and Instrumental variables. (15 lectures)

Reference Books:

1. “Introductory Econometrics: A Modern Approach” by Jeffrey Wooldridge, International Edition, Cengage
2. “Econometric Analysis of Cross-section and Panel Data” by Jeffrey Wooldridge, MIT Press, 2010
3. “Econometric Analysis” by William Greene, Pearson, 2018
4. Kennedy, P., *Guide to Econometrics* (6th ed.) MIT Press
5. Kmenta Jan, “Elements of Econometrics”, Macmillan

Title	**Economic Challenges and Data- driven Approach**	**Number**	7XXX
Department	School of Liberal Arts	Structure (L- T-P-C)	3-0-0 [3]
Offered for	UG, PG, PhD	Type	Elective
Prerequisite/ Corequisite			

Objectives:

- The economic growth and development trajectory is vulnerable to business cycles, which, according to mainstream economics, requires government intervention from time to time. These interventions are more large-scale when faced with bigger events such as recessions and financial crises. Why these vulnerabilities exist and why and these events occur, will be explored, along with the forms that corrective interventions.
- The currently dominant economic system has led to vulnerabilities and challenges such as obstacles to structural transformation, unemployment, inequality, poverty, high levels of informality in the economy. Why this is the case and what policies (if at all) may address these challenges will be explored.
- The fourth industrial revolution, characterised by increasing automation and ICT, presents challenges for production structures and capital-labour relations, and innovations such as Cryptocurrency, as a decentralised medium of payment, have both potential and risks in terms of evasion of a central monetary authority whose job is to stabilise the economy.
- All the above larger set of challenges has attracted the attention of empirical research by economists, which uses Econometric methods as well as economic policy research, such as impact evaluations using Randomised Controlled Trials. The basic rationale of these approaches and some of the research findings will be introduced.

Learning Outcomes:

- Contextualise mathematical and computational models in contemporary economic challenges (some of which have lessons learned or unlearnt from the past), which themselves can be contextualised in a spectrum of macroeconomic as well as microeconomic schools of thought, in both of which the spectral units are often in opposition to each other.
- Be equipped with a critical thinking outlook and the knowledge foundation, based on which creative econometric models can be built to address real-world challenges

- Have a basic preview of modern economists' research approach and methodology, which is primarily econometric, whether based on primary data, secondary data or experimental or quasi-experimental data

Contents:

1. Economic growth, business cycle and financial crisis (7 lectures)
2. Structural Transformation: Agricultural productivity and labour absorption in industry and services. Employment: Occupational structure in the organised and unorganised (or informal) sectors; open-, under and disguised- unemployment (rural and urban); employment schemes and their impact (7 lectures)
3. Fourth Industrial Revolution: Increasing ICT and AI, consequences for labour-capital relations, Future of work, Cryptocurrency (7 lectures)
4. Poverty and Inequality definitions, measures and mechanisms: Inequality axioms; comparison of commonly used inequality measures; connections between inequality and development; poverty measurement; characteristics of the poor; mechanisms that generate poverty traps, and path dependence of growth processes (7 lectures)
5. Market Failures and policy: Asymmetric information, moral hazard and their manifestations in land, labour and credit markets, informal institutions (7 lectures)
6. Addressing the above economic challenges: basic rationale of econometric approaches for research, their findings and implications (7 lectures)

Selected references

1. Angrist, Joshua D., and Jörn-Steffen Pischke. Mastering 'Metrics: The Path from Cause to Effect. Princeton University Press, 2014.
2. Banerjee, Abhijit V., and Esther Duflo. Poor economics: Rethinking poverty & the ways to end it.
Random House India, 2011.
3. Banga, R. (2014). Trade facilitation and `hollowing-out' of Indian manufacturing. *Economic and Political Weekly,* 49(40), 57-63.
4. Basole, A., Basu, D., Bhattacharya, R. (2015). Determinants and Impacts of Subcontracting: Evidence from India's Unorganised Manufacturing Sector. *International Review of Applied Economics,* 29, 374-402.
5. Duflo, Esther, and Abhijit Banerjee. *Poor Economics*. Vol. 619.
Public Affairs, 2011. Dreze, J., Sen, A. (2013). *India: An Uncertain Glory*.
Allen Lane.
6. Fama, Marco, Andrea Fumagalli, and Stefano Lucarelli. "Cryptocurrencies, monetary policy, and new forms of monetary sovereignty." *International Journal of Political Economy* 48.2 (2019): 174-194.
7. Gulati, A., Saini, S. (2017). 25 years of Policy Tinkering in Agriculture. In R. Mohan (Ed.): *India Transformed: 25 Years of Economic Reforms.* Penguin.
8. Mehrotra, S. (2015). *Realising the Demographic Dividend: Policies to Achieve Inclusive Growth in India.* Cambridge University Press.
9. Piketty, T., Saez, E. (2014). Inequality in the Long Run, *Science,* 344.
10. Rajan, R. (2011). *Fault lines: How Hidden Fractures Still Threaten the World Economy*. Princeton University Press.
11. Ray, Debraj. *Development Economics*. Princeton University Press, 1998.
12. Rodrik, D. (2011). The Globalisation Paradox: *Why Global Markets, States, and Democracy cannot coexist*. Oxford University Press.
13. Sen, K., Das, D. (2015). Where have all the Workers Gone? The Puzzle of Declining Labour Intensity in Organised Indian Manufacturing. *Economic and Political Weekly,* 50(23), 108-115.
14. Varoufakis, Y., Halevi, J., & Theocarakis, N. (2012). Modern Political Economics: Making Sense of the Post- 2008 World. *Routledge.*

Title	**Health Policy, Systems, and Financing**	Number	LAL7510
Department	Humanities and Social Sciences	L-T-P [C]	3-0-0[3]
Offered for	UG and PG	Type	Elective
Prerequisite	Nil		

Objectives

1. Introduce critical and comparative understanding of the theory and practice of health policy with special focus on India.
2. Facilitate students to understand components of health systems and its financing mechanism.
3. Introduce different national level secondary health data sets and its relevance in research and policy making.

Learning Outcomes

1. Understand the importance of health policy and how to define it, with a special focus on India.
2. Provide a brief overview of the evolution of Indian health policy since Independence.
3. Understand the different components of health systems and their financing mechanisms.
4. Identify available national-level secondary datasets in the health sector and their relevance to research and policy-making.

Contents

Making Health Policy: The health policy framework: context, process, and actors; power and policy process; the state and private sector in health policy; agenda setting; government and policy process; policy implementation and evaluation *(10 lectures)*.

Evolution of India's Health Policy: Bhore Committee-1946; National Health Policy-1983, 2002, and 2017; Alma-Ata Declaration-1978; National Rural Health Mission-2005; Ayushman Bharat Yojana-2018 *(8 lectures).*

Understanding Health System: Service Delivery; health workforce; health information system; medical product; vaccines; and technology; Financing; leadership and governance *(7 lectures)*.

Healthcare financing: Health transition, disease burden, and health expenditure pattern; collecting revenue, pooling risk, and purchasing services; risk pooling mechanism; external assistance for health; improving health outcome *(8 lectures)*.

National Level Secondary Health Data Sets: National Sample Survey; National Family Health Survey; Census; Longitudinal Ageing Study in India *(8 lectures).*

Textbook

1. Buse, K., Mays, N., & Walt, G. (2012). Making health policy. McGraw-Hill Education (UK).
2. World Health Organisation. (2007). Everybody's business-strengthening health systems to improve health outcomes: WHO's framework for action.
3. Gottret, P., & Schieber, G. (2006). Health financing revisited: a practitioner's guide. The World Bank.

Reference Books/ Re

1. Blank, R., Burau, V., & Kuhlmann, E. (2017). Comparative health policy. Macmillan International Higher Education.
2. Mills, A., Bennett, S., & Russell, S. (2001). The challenge of health sector reform: What must governments do? New York: St.
3. Ministry of Health and Family Welfare, Government of India (2017). National Health Policy - 2017. New Delhi
4. Government of India (1946) 'Report of the Health Survey and Development Committee (Bhore Committee) IV. Manager of Publications, Delhi.

Online Course Materials

Emanuel E.J. Economics of Healthcare Delivery. University of Pennsylvania: https://www.who.int/health_financing/training/e-learning-course-on-health-financing-policy-for-uhc/en/

Title	**Global Health**	Number	LAL7010
Department	Humanities and Social Sciences	L-T-P [C]	3-0-0 [3]
Offered for	UG and PG	Type	Elective
Prerequisite	Nil		

Objectives

1. To introduce basic concepts of global health and its relevance globally, with special reference to lower and middle-income countries (LMICs).
2. To understand the various determinants of global health and policy options to address global public health issues.
3. To examine how medical technologies should be used to improve health in low and middle-income countries.

Learning Outcomes

1. Understand the basic concepts of global health and their relevance to policy-making.
2. Analyse the role of various determinants on global health.
3. Understand the role of medical technologies in improving health.

Contents

Principles, Measurements, and Health-Development links: The principles and goals of global health; Health determinants, measurements and status of health globally; The global burden of diseases; Health, Education, equity and economy (8 lectures).

Cross-Cutting Global Health Themes: Ethical and human rights concerns in global health; Culture and health; Health systems (8 lectures).

Burden of Disease: The environment and health; diseases of poverty, Nutrition and global health; Women's health; Communicable and Non-communicable diseases; Mental health; injuries; case studies from South-East Asian countries (12 lectures).

Global Health Institutions: Role of WHO over the period of time; World Bank; UNICEF; USAID; European commissions (5 lectures).

Role of Technologies in Improving Global Health: Global health needs and contribution of technologies; Frugal technologies; Implementation and innovation in technology for health; Disruptive technologies for healthcare in high-income countries; Equity and universal coverage of technology (7 lectures).

Text Books:

1. Richard Skolnik (2020), Global Health 101. Johns Bartlett Learnings.

2. Birn, A. E., Pillay, Y., & Holtz, T. H. (2017). Textbook of global health. Oxford University Press.

Reference Books/Reports:

1. Detels, R. (Ed.). (2015). *Oxford textbook of global public health* (Vol. 2). Oxford Textbook.
2. Glassman, A., & Temin, M. (2016). *Millions saved: New cases of proven success in global health.* Brookings Institution Press.
3. Farmer, P., Kim, J. Y., Kleinman, A., & Basilico, M. (2013). *Reimagining global health: an introduction* (Vol. 26). Univ of California Press.
4. Jamison, D. T., Summers, L. H., Alleyne, G., Arrow, K. J., Berkley, S., Binagwaho, A., ... & Yamey, G. (2013). Global health 2035: a world converging within a generation. *The Lancet*, 382(9908), 1898-1955.
5. Watkins, D. A., Yamey, G., Schäferhoff, M., Adeyi, O., Alleyne, G., Alwan, A., ... & Summers, L. H. (2018). Alma-Ata at 40 years: reflections from the Lancet Commission on Investing in Health. *The Lancet,* 392(10156), 1434-1460.
6. Howitt, P., Darzi, A., Yang, G. Z., Ashrafian, H., Atun, R., Barlow, J., ... & Wilson, E. (2012). Technologies for global health. *The Lancet*, *380*(9840), 507-535.

Online Course Material

1. Skolnik R. Essentials of Global Health, Yale University. https://www.coursera.org/learn/essentials-global-health

Course Title	Indian Knowledge Systems (IKS)	Course No.	LAL7960
Department/School	SoLA	Structure (L-T-P) [C]	3-0-0 [3]
Offered for	PhD	Type	Elective
Pre-requisite		None	

Objectives

- To introduce learners to the foundational concepts of Indian Knowledge Systems.
- To explore the contributions of ancient and medieval India in science, mathematics, astronomy, medicine, art, architecture, language, and philosophy.
- To understand the contemporary relevance and applications of IKS.

Learning Outcomes

The students will be able to

- Comprehend the fundamental principles and philosophical underpinnings of Indian Knowledge Systems.
- Develop a multidisciplinary perspective on IKS covering areas such as science, technology, mathematics, astronomy, medicine, architecture, linguistics, and art.
- Critically examine how ancient Indian knowledge can inform modern education, research, sustainability, health, and technology.

Contents

Introduction to IKS - Meaning and scope of IKS, Importance in the NEP 2020 and national initiatives, Overview of primary sources (*Vedas*, *Vedāṅgas*, *Upaniṣads*, *Purāṇas*, *Darśana* etc.) (**5 lectures**)

Mathematics in India - Overview of Indian mathematicians and their works and their contribution (**8 lectures**)

Astronomy and Timekeeping – *Jyotiṣa* (*Siddhānta, Saṃhitā, Horā*), Indian calendar systems, Role of astronomy in ritual, agriculture, and architecture (**8 lectures**)

***Āyurveda* and *Yoga*-** Fundamental principles (*Doṣa*, *Dhātu*, *Mala)*, *Āyurveda* and *Yoga* texts (*Carakasaṃhitā*, *Suśrutasaṃhitā*, *Aṣṭāṅgahṛdaya, patañjaliyogasūtra*) (**6 lectures**)

Language and Literature - Fundamental Pāṇinian grammar, Introduction of *Kāvyas,* types of *Nāṭakas.* (**6 lectures)**

Indian Society, Polity and Economy - *Arthaśāstra*, *Dharmaśāstra*, *Smṛtis,* Indigenous educational systems (*Gurukulas*, *Pāṭhaśālās*), Oral tradition, Role of women in IKS. (**6 lectures**)

Text Books

- S. Balachandra Rao, *Indian Astronomy: An Introduction*, Hyderabad 2000.
- D. M. Bose, S. N. Sen, and B. V. Subbarayappa, *A concise history of Science in India*, INSA, New Delhi, 1971.
- Tilak S. Rao, अष्टाध्यायीप्रवेशः (*Aṣṭādhyāyīpraveśaḥ*), 2 vols, Veda Vijnana Shodha Samsthanam, Bengaluru, 2017.

Reference Books

- B. Datta and A. N. Singh, *History of Hindu Mathematics*, 2 vols, Lahore 1935, 1938 (repr. Asia Publishing House, Bombay 1962; repr. Bharatiya Kala Prakashan, Delhi 2004).
- Priyadaranjan Ray and S. N. Sen, *The cultural heritage of India*, The Ramakrishna Mission Institute of Culture, Calcutta, 1937.

Online Reading Materials

- https://swayam.gov.in/

Comments of National Advisory Committee

The School of Liberal Arts constituted a National Advisory Committee (NAC) to obtain feedback on the MSc CSS and MSc DH programmes. The competent authority approved the following members of the NAC for SoLA. The first meeting of the NAC was held on Wednesday, 20 May 2026. The Committee provided the following observations and recommendations:

- The Committee appreciated the overall design of the programmes and the interdisciplinary nature of the courses offered.
- Members suggested incorporating themes related to ethics, privacy, and the digital domain within the curriculum.
- The Committee also recommended the inclusion of a linguistics component in the course Perspectives on Language and Literary Studies.
- Greater emphasis should be placed on AI Ethics within the curriculum.
- Soft skills components/courses, along with modules on data regulation and governance, should be introduced.
- The curriculum should include computational tools and programming languages such as Python and R.
- The programmes should maintain a stronger focus on the Social Sciences, including the integration of social theory perspectives.
- Opportunities for introducing an integrated MSc–PhD dual degree programme may be explored in the future.